

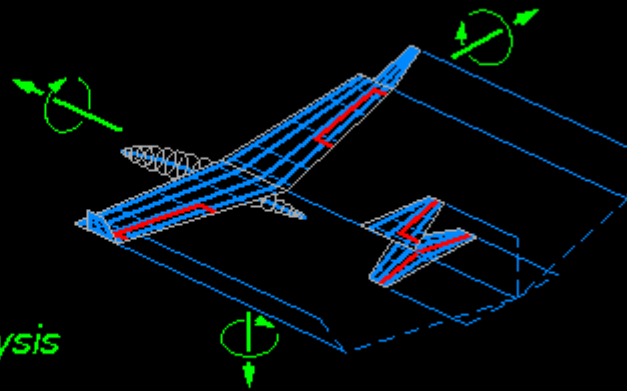
AVL

Aerodynamic Analysis

Trim Calculation

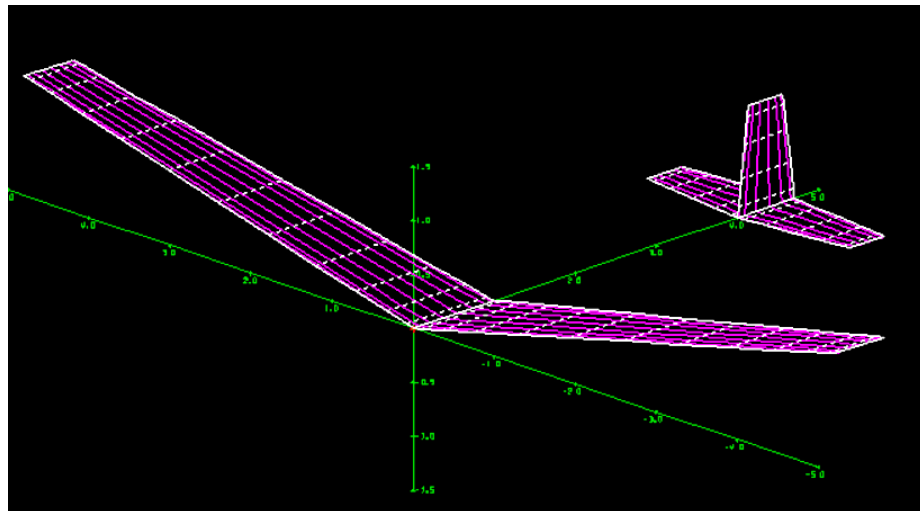
Dynamic Stability Analysis

Aircraft Configuration Development



Using AVL

Ney Secco





I HAVE NO IDEA

WHAT I'M DOING

quickmeme.com

Introduction

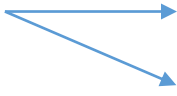
- AVL uses the Vortex Lattice Method (VLM).
- This method solves the Potential Flow equation.
 - No viscosity
 - No compressibility

- AVL only gives induced drag.

$$C_D = C_{D0} + C_{Dw} + C_{Dind}$$

- Executable, manual and examples can be found at:
 - <http://web.mit.edu/drela/Public/web/avl/>

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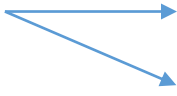
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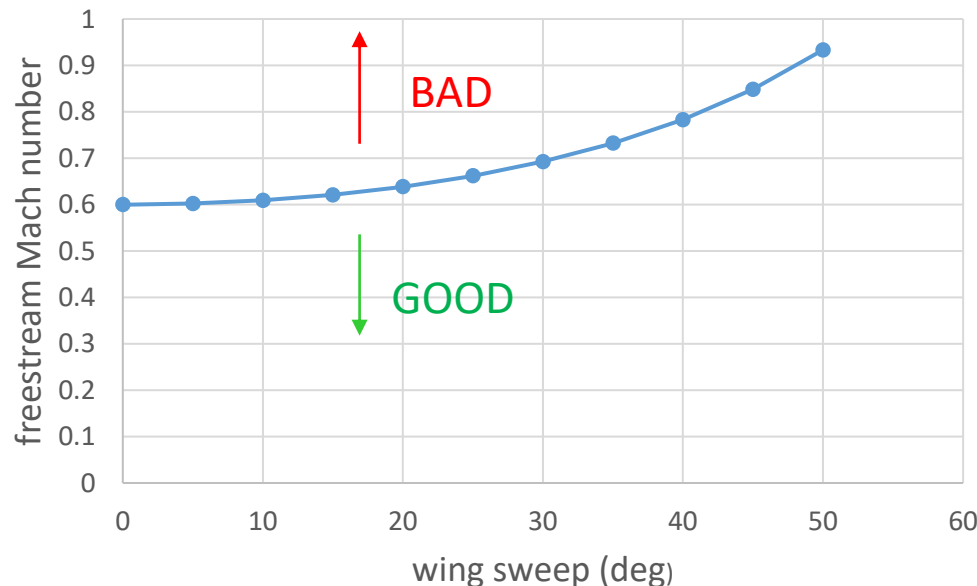
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 - No friction drag
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 - No compressibility 
 - No wave drag
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- $$C_D = C_{D0} + C_{Dw} + C_{Dind}$$
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Introduction

- AVL uses Prandtl-Glauert corrections for compressibility effects
- This correction is reasonable when the perpendicular Mach number is below 0.6

$$M_{perp} = M \cdot \cos(\lambda) < 0.6$$

freestream Mach number wing sweep



Pre-requisites

- In order to use AVL correctly you will need:
 - Aircraft geometry (three views, including control surfaces)
 - Airfoil coordinates (see sd7037.dat file)
 - CG range of the entire airplane
 - Takeoff, cruise and landing lift coefficients

$$C_{L,TO} = \frac{C_{Lmax,TO}}{1.2^2} \quad C_{L,CR} = \frac{2W_{CR}}{\rho_{CR}V_{CR}^2S_{ref}} \quad C_{L,LD} = \frac{C_{Lmax,LD}}{1.3^2}$$

Input file

- In order to run an aerodynamic analysis, we need to write an input file with extension “.avl” that describes the aircraft geometry.
- You can use any text editor (e.g. Notepad) to write this file.
- Lines starting with “#” are comments.
- For now we will not need “.mass” files.
- In this tutorial we will use the file “test.avl”, which is available on Ctools.

Input file - Header

Plane Vanilla → Airplane name

#Mach → Freestream Mach number
(sometimes it does not work, so you have to modify it at runtime)

0.0

#IYsym IZsym Zsym → Creates symmetry planes
(useful to simulate ground effect)

0 0 0.0

#Sref Cref Bref

8.0 1.0 10.0

#Xref Yref Zref → Reference quantities

0.50 0.0 0.0

#CDp → Parasite drag

0.0150

CG position (THIS IS VERY IMPORTANT)!!!

Input file – Wing definition

SURFACE

Wing

#Nchordwise Cspace Nspanwise Sspace

8 1.0 12 1.0 → Discretization parameters

YDUPLICATE

0.0 → This mirrors the surface using Y=0 as symmetry plane

ANGLE

6.0 → Incidence angle – tilts the entire surface

COMPONENT

1 → Always use this so that different surfaces interact

#-----

SECTION → Root section definition

#Xle Yle Zle Chord Ainc Nspanwise Sspace

0. 0. 0. 1.0 0.0 0 0 → Section leading edge coordinates

AFILE

sd7037.dat

CDCL

-0.1570 0.01396 0.3702 0.00545 1.4721 0.02924

Airfoil drag polar (represented by 3 CL-CD pairs)

ATTENTION: Do not use this if you already added the wing parasite drag in the CDp parameter

CLAF

1.0 → $CLAF = (2D \text{ lift slope}) / (2 * \pi)$

CONTROL

#Cname Cgain Xhinge HingeVec SgnDup

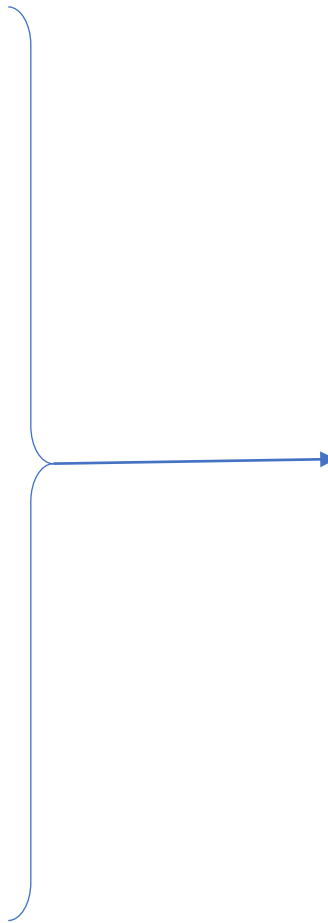
aileron -1.0 0.75 0.0 0.0 0.0 -1.0

#-----

Control surface definition
(use negative Xhinge to model slats)

Input file – Wing definition

```
#-----  
SECTION  
#Xle Yle Zle Chord Ainc Nspanwise Sspace  
0.2 5.0 1.0 0.6 0.0 0 0  
AFILE  
sd7037.dat  
CDCL  
-0.1570 0.01396 0.3702 0.00545 1.4721 0.02924  
CLAF  
1.0  
CONTROL  
#Cname Cgain Xhinge HingeVec SgnDup  
aileron -1.0 0.75 0.0 0.0 0.0 -1.0
```



Tip section definition

Input file – HT definition

```
#=====
```

```
SURFACE
```

```
H-stab
```

Discretization parameters

```
#Nchordwise Cspace Nspanwise Sspace
```

```
6      1.0  6      1.0
```

```
YDUPLICATE
```

```
0.0
```

Mirror surface

```
TRANSLATE
```

```
4.0 0.0 0.0
```

```
COMPONENT
```

```
1
```

```
##-----
```

```
SECTION
```

```
#Xle Yle Zle Chord Ainc Nspanwise Sspace
```

```
0.0  0.0  0.0  0.7  0.  0      0
```

```
CDCL
```

```
-1.0298 0.04048 0.0 0.00530 1.0298 0.04048
```

```
#Cname Cgain Xhinge HingeVec SgnDup
```

```
CONTROL
```

Control surface definition

```
elevator 1.0  0.7  0.0 1.0 0.0 1.0
```

```
#-----
```

```
SECTION
```

```
#Xle Yle Zle Chord Ainc Nspanwise Sspace
```

```
0.14  1.25  0.0  0.42  0.  0      0
```

```
CDCL
```

```
-1.0298 0.04048 0.0 0.00530 1.0298 0.04048
```

```
#Cname Cgain Xhinge HingeVec SgnDup
```

```
CONTROL
```

```
elevator 1.0  0.7  0.0 1.0 0.0 1.0
```

Tip section definition

Input file – VT definition

```
##=====
```

```
SURFACE
```

```
V-stab
```

```
#Nchordwise Cspace Nspanwise Sspace
```

```
6      1.0   5      1.0
```

Discretization parameters

```
TRANSLATE
```

```
4.0 0.0 0.0
```

```
COMPONENT
```

```
1
```

```
#-----
```

```
SECTION
```

```
#Xle Yle Zle Chord Ainc Nspanwise Sspace
```

```
0.0  0.  0.0  0.7  0.  0      0
```

```
CDCL
```

```
-1.0298 0.04048 0.0 0.00530 1.0298 0.04048
```

```
#Cname Cgain Xhinge HingeVec SgnDup
```

```
CONTROL
```

```
rudder 1.0  0.5  0.0 0.0 1.0 1.0
```

Control surface definition

```
#-----
```

```
SECTION
```

```
#Xle Yle Zle Chord Ainc Nspanwise Sspace
```

```
0.14  0.  1.0  0.42  0.  0      0
```

```
CDCL
```

```
-1.0298 0.04048 0.0 0.00530 1.0298 0.04048
```

```
#Cname Cgain Xhinge HingeVec SgnDup
```

```
CONTROL
```

```
rudder 1.0  0.5  0.0 0.0 1.0 1.0
```

Tip section definition

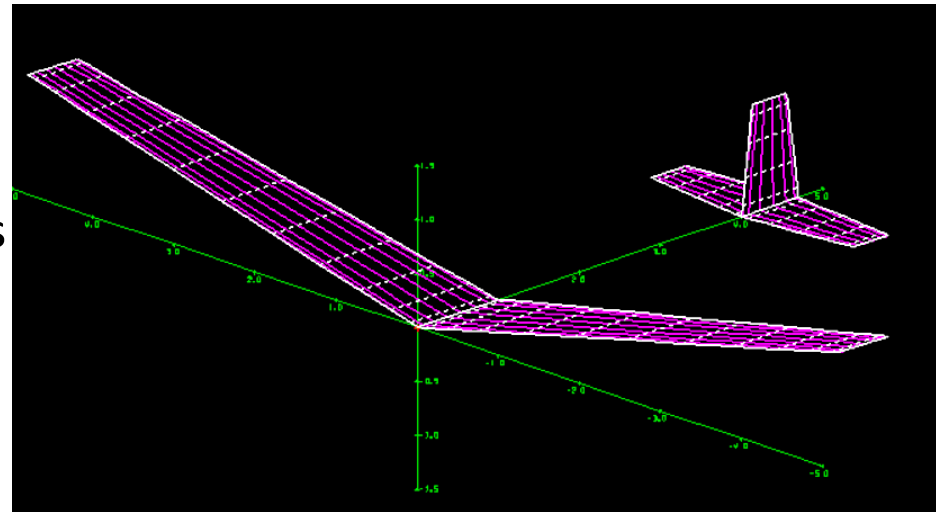
```
#-----
```

IMPORTANT!

Never use YDUPLICATE for surfaces that lie on the $y=0$ plane.

Checking your geometry

- First you should check if your geometry file is correct!
- You can visualize your geometry with the following commands (after you open AVL):
 - load test.avl
 - oper
 - g
 - k
 - (use the L,U,D,R keyboard keys to rotate the geometry and press space when finished).



Changing Mach number

- Sometimes AVL does not read the Mach number specified by the input file.
- You can run the following commands to modify the Mach number:
 - load test.avl
 - oper
 - m
 - mn
 - 0.4
 - <enter>
- You can verify the Mach number while checking results.

Things you can do with AVL

- Compute static margin
- Predict stall position
- Get trimmed polar
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- Stability derivatives

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Static margin

- Run the following commands:
 - load test.avl
 - oper
 - x
 - st
 - (you can select a file name or just press enter)
- Read the neutral point position (X_{np}) on the screen

```
Neutral point  Xnp =  0.662346
```

- Compute static margin

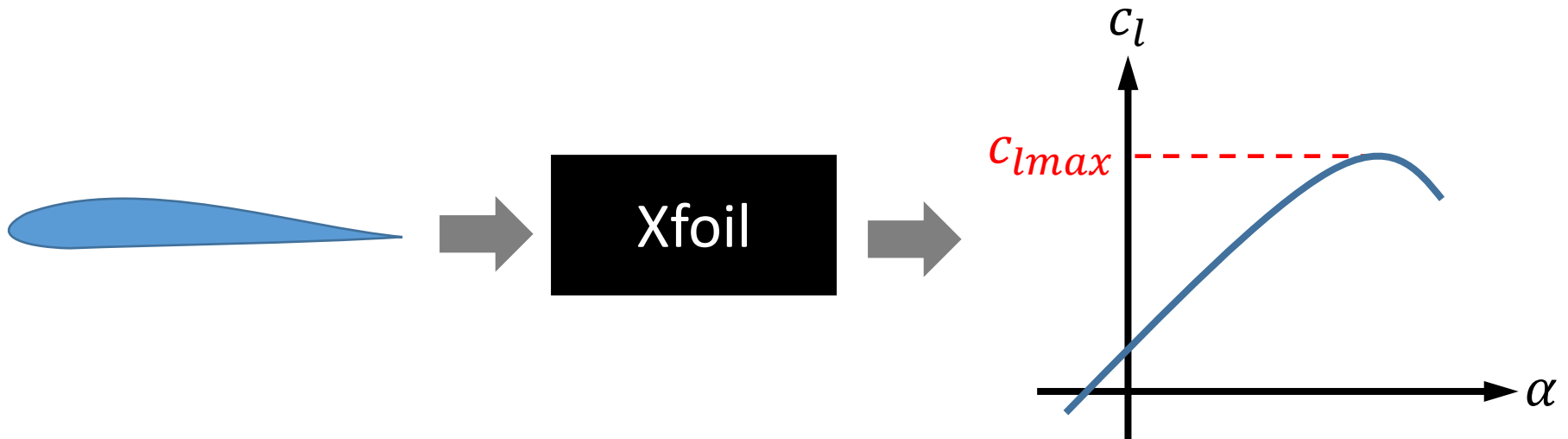
$$SM = \frac{x_{NP} - x_{CG}}{MAC}$$

Things you can do with AVL

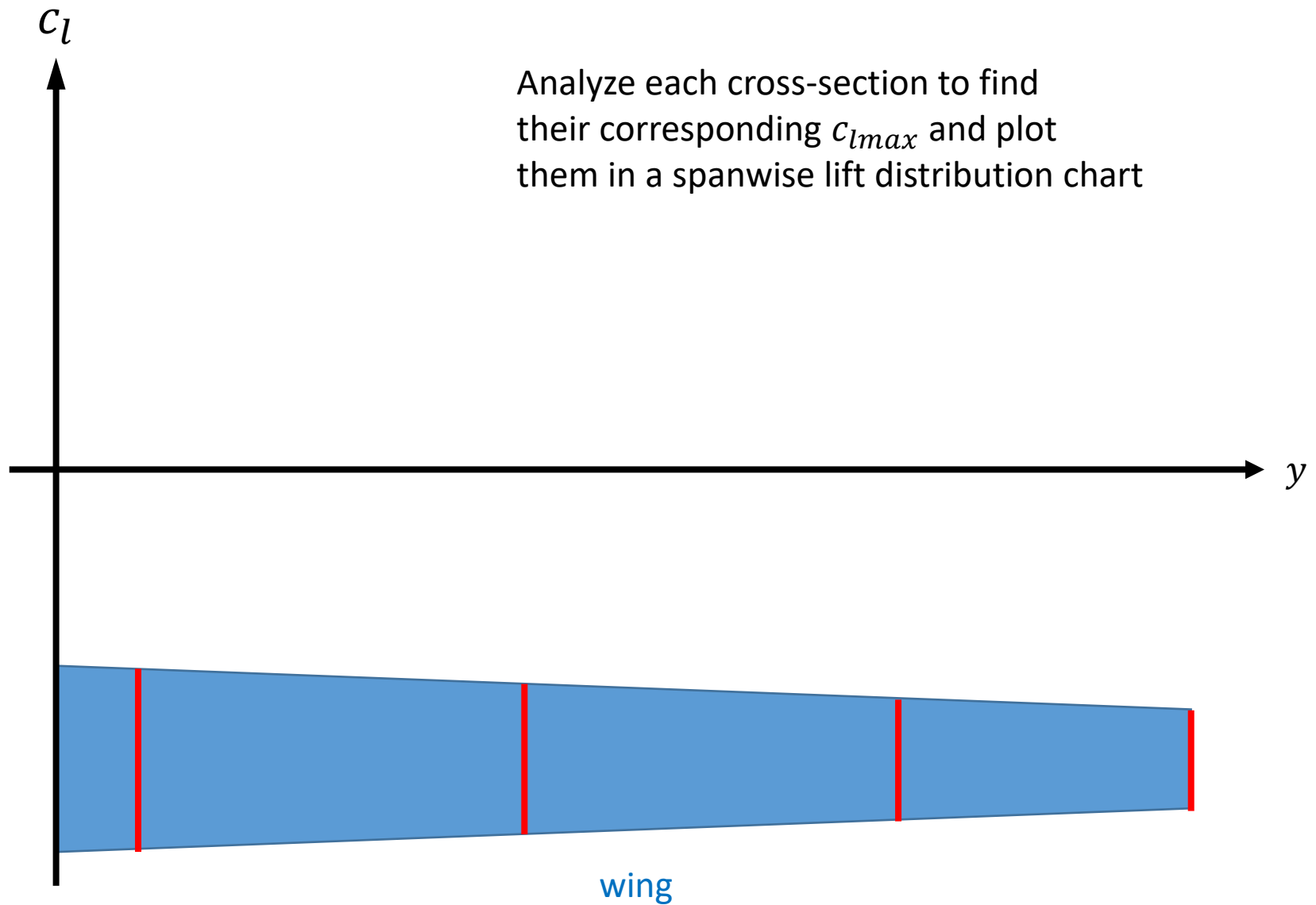
- Compute static margin
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Critical Section Method

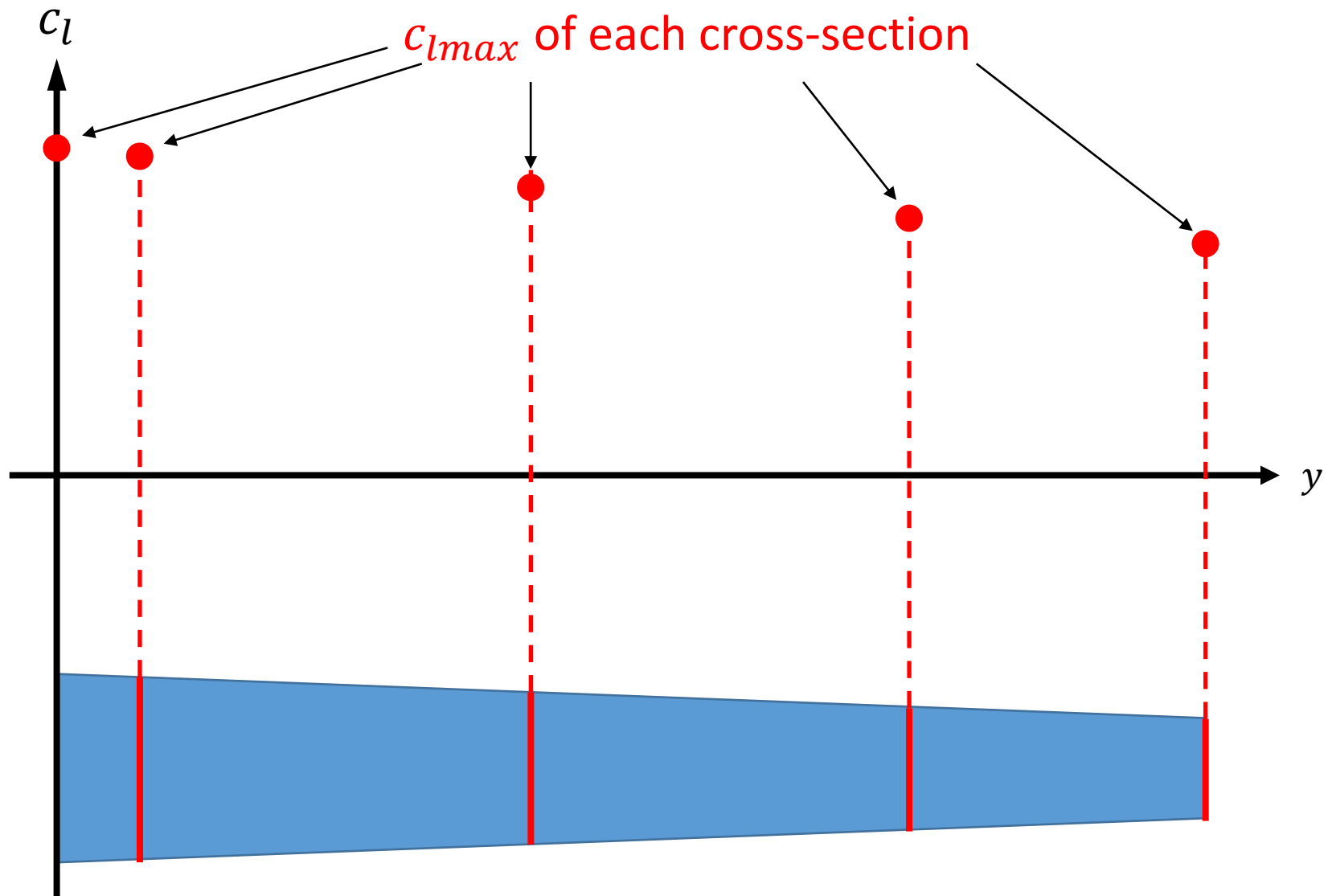
- Simple method to estimate stall onset
- You will need the 2D c_{lmax} of the wing cross-sections
- You can use Xfoil or experimental data to get c_{lmax}



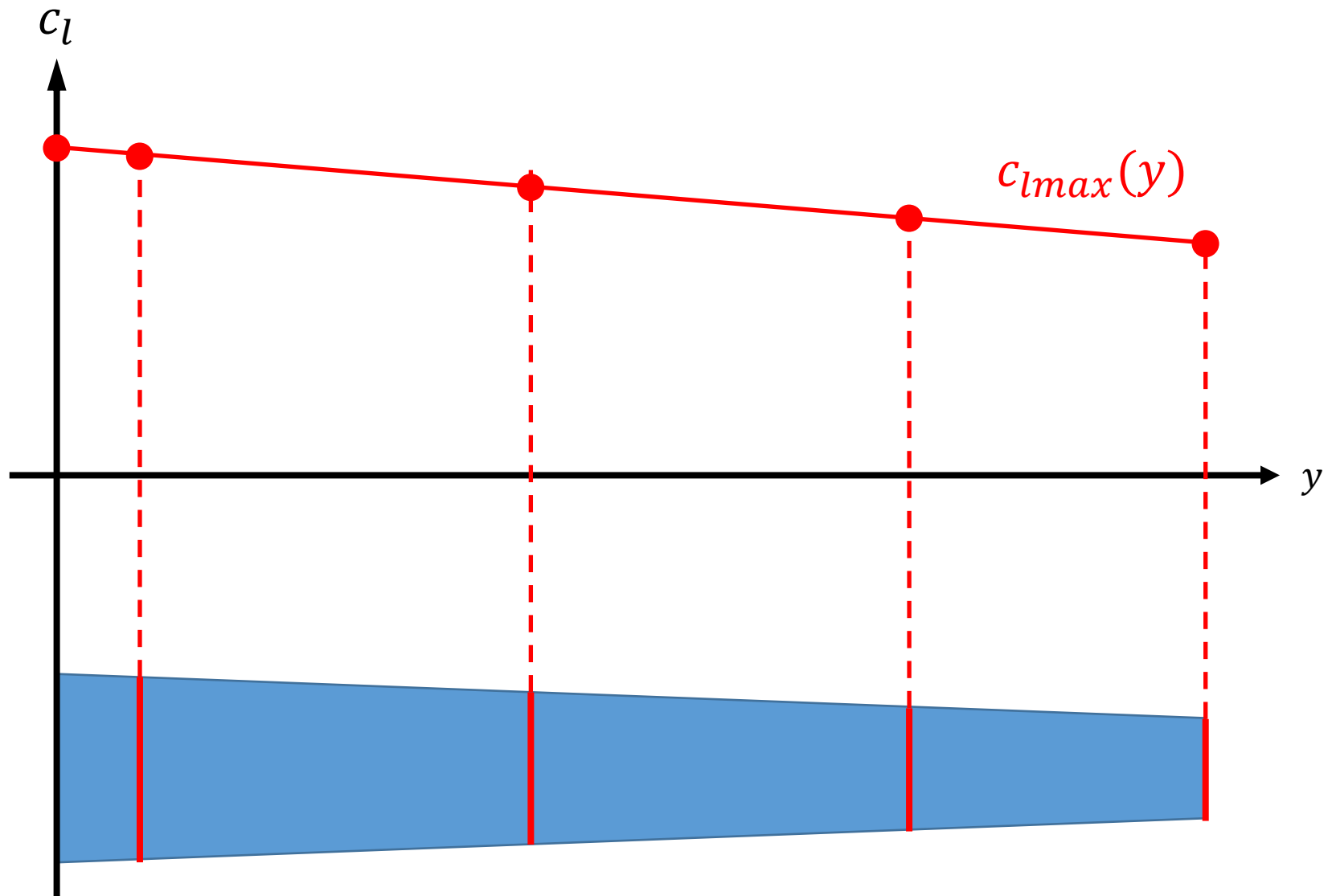
Critical Section Method



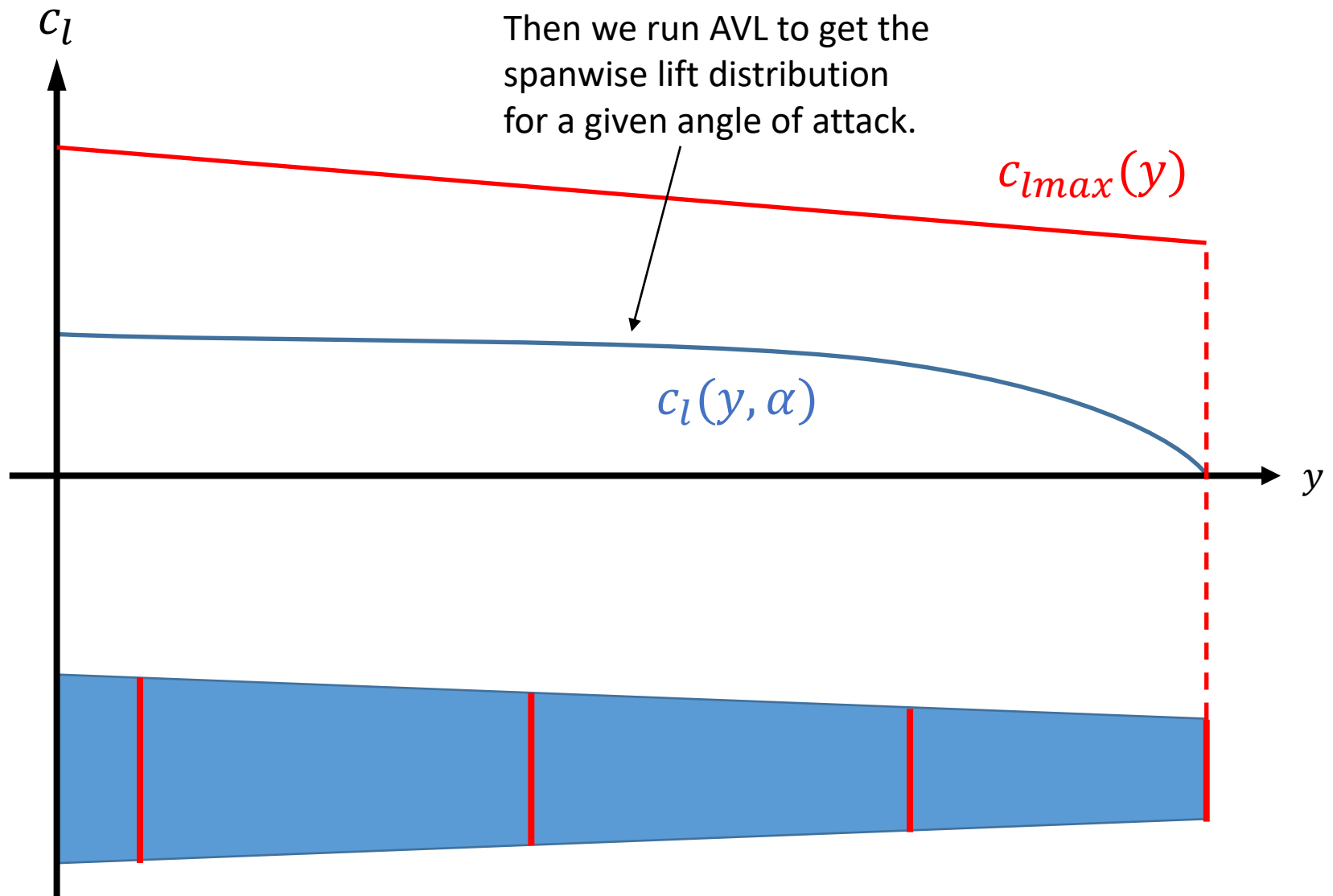
Critical Section Method



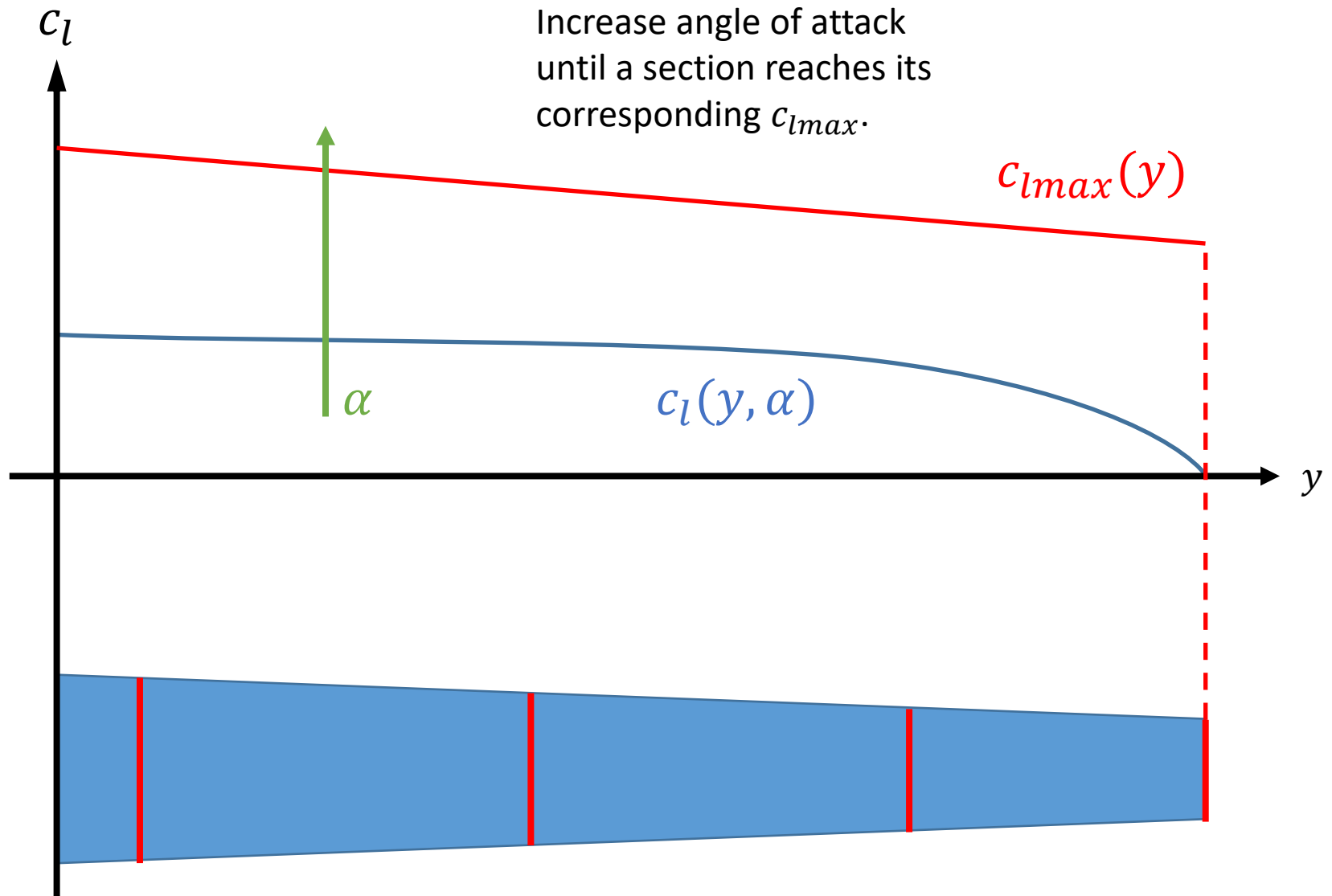
Critical Section Method



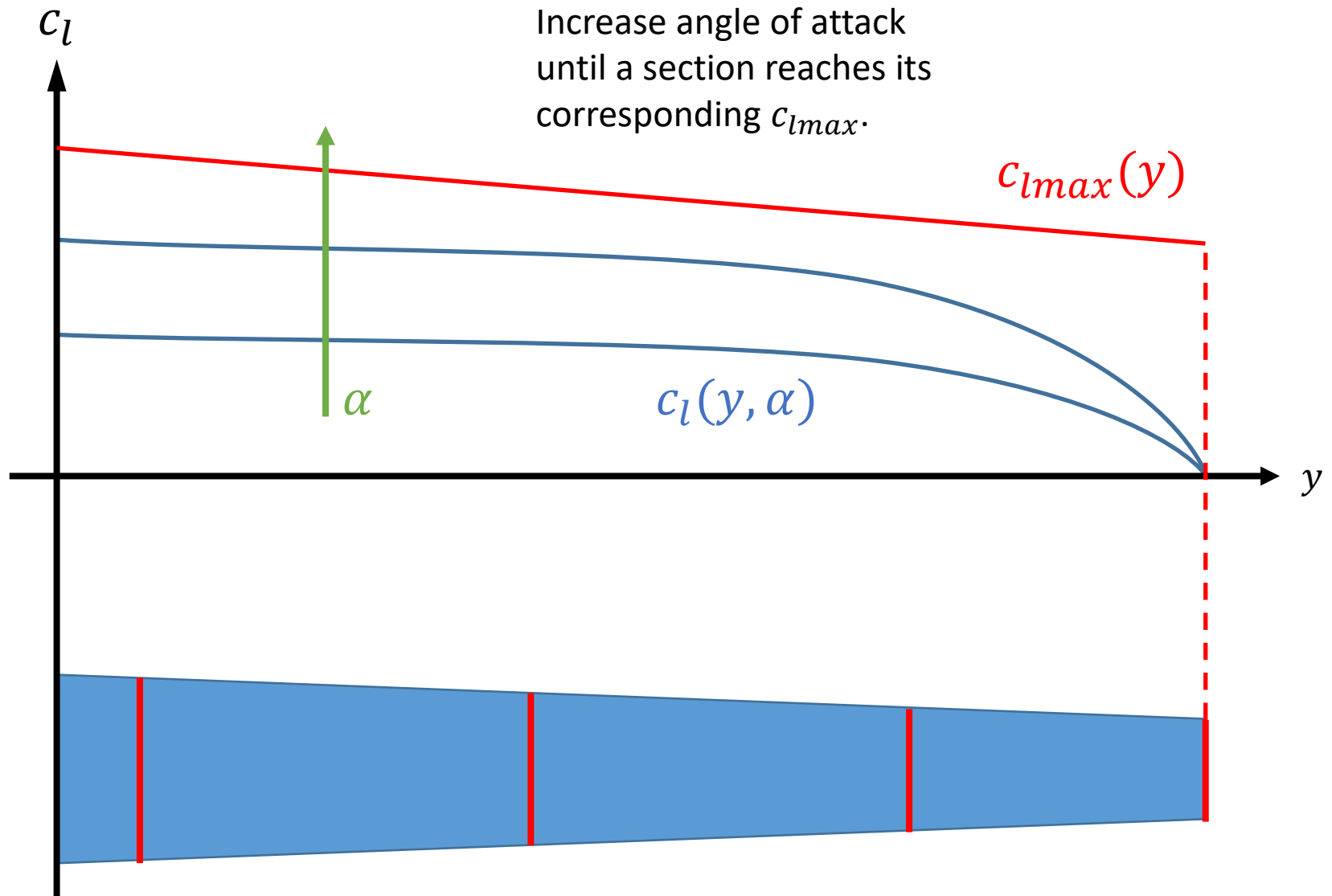
Critical Section Method



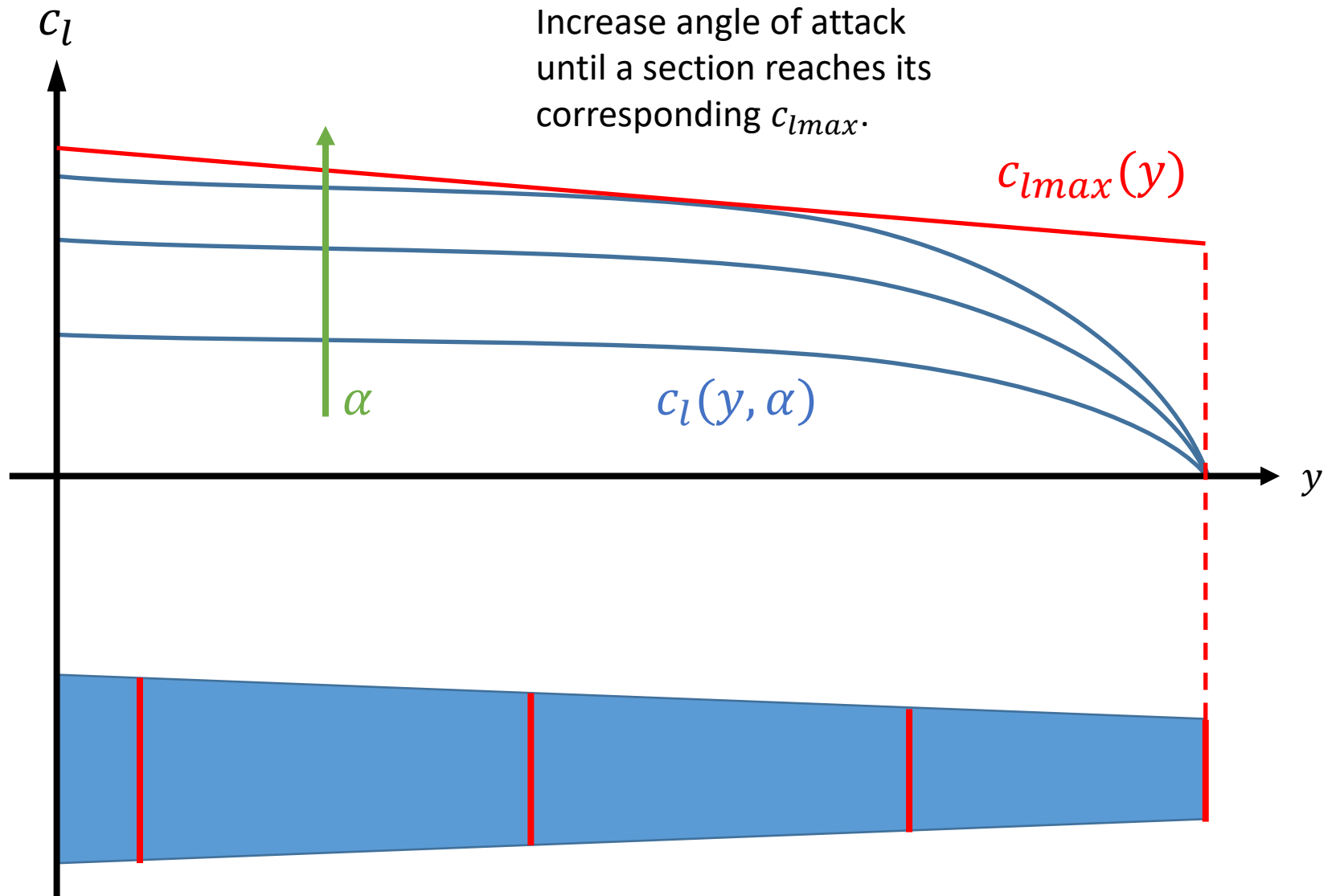
Critical Section Method



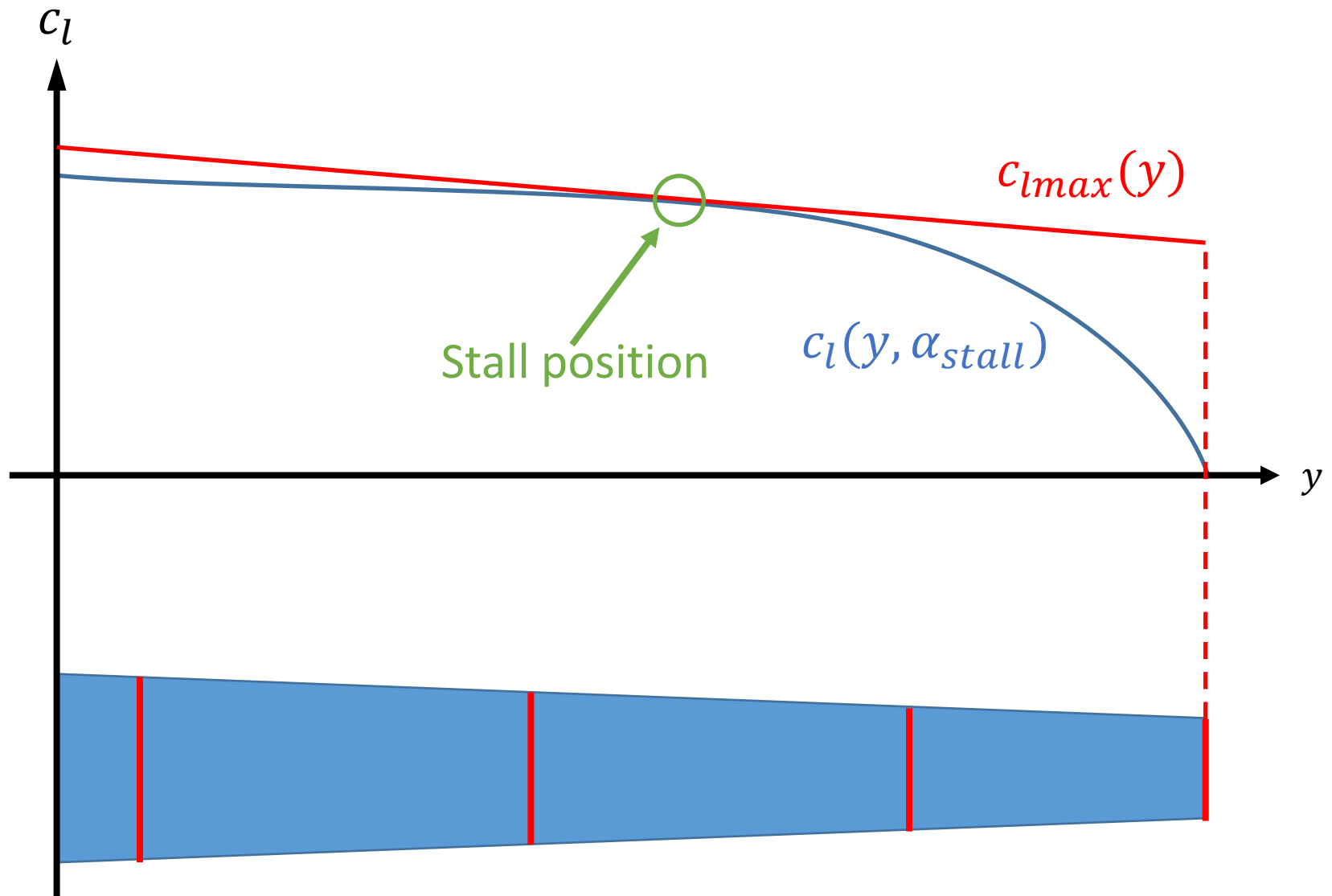
Critical Section Method



Critical Section Method



Critical Section Method

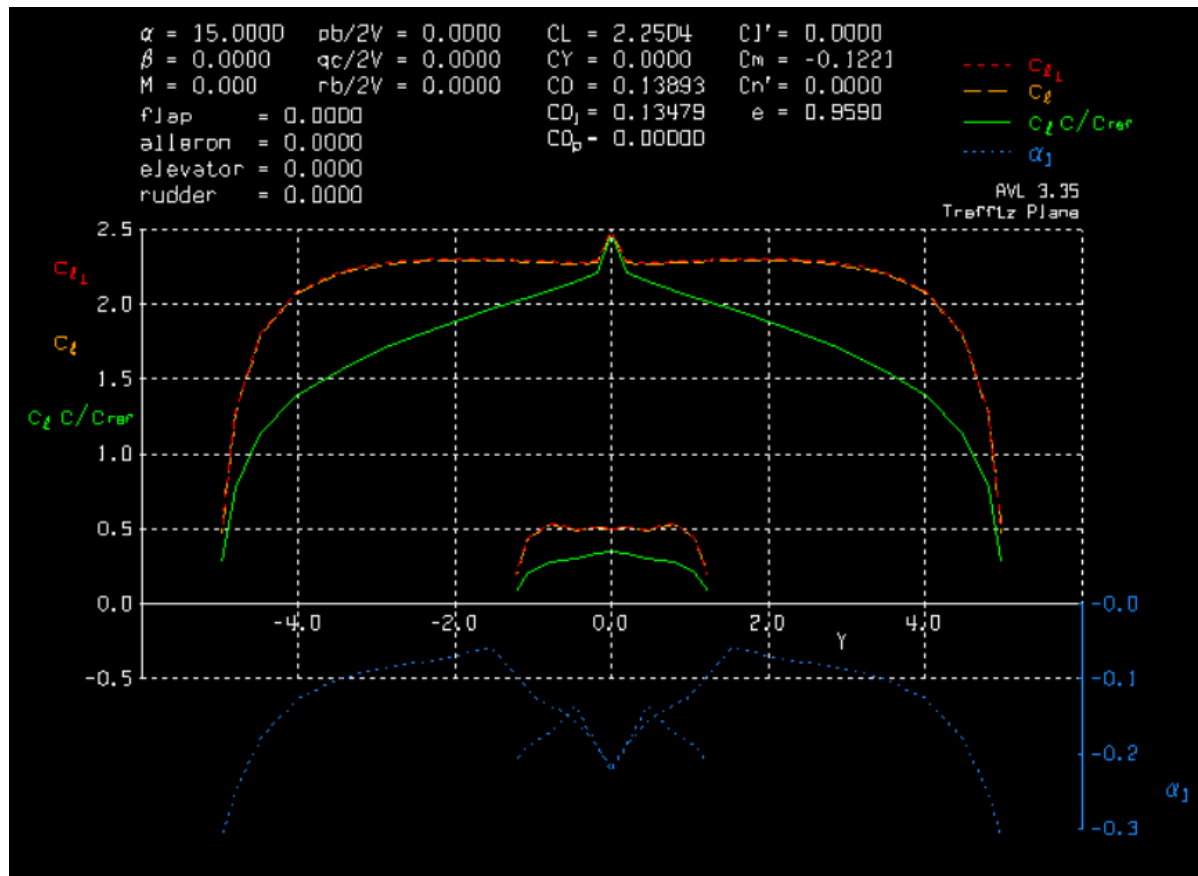


Predicting stall position

- It is important to ensure that the wing stall does not begin near the wingtip and ailerons.
- We can use the c_l distribution to estimate where stall begins.
- Run the following commands:
 - Load test.avl
 - oper
 - a a 10
 - x
 - t
 - (now look for the peak of the red curve)
- The peak of the red curve can be used as an estimate of where the stall begins for high-AR high-sweep wings. This peak should not be in the aileron region.

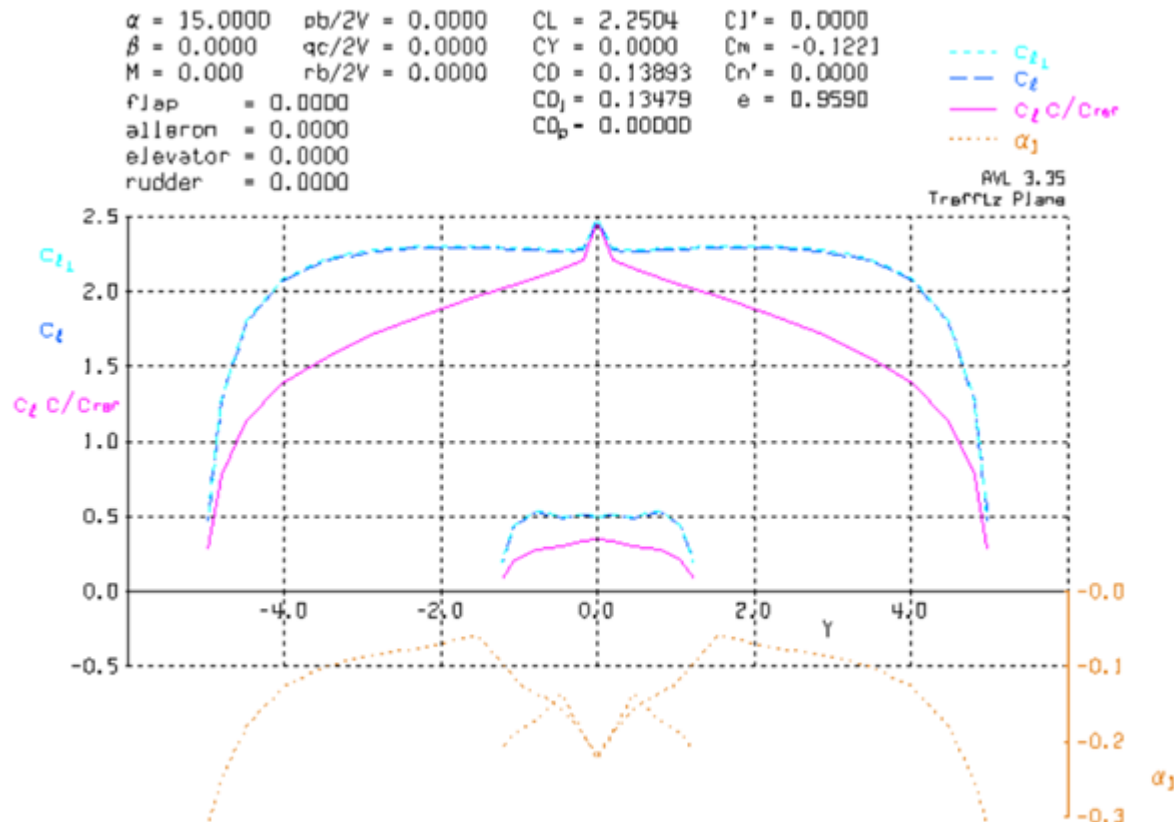
Predicting stall position

- You can twist the wing (washout) to reduce the effective AoA of the tip sections and shift the C_L peak towards the root.



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Trimmed polar

- We need to get values of C_L and C_D for $C_M = 0$.
- We can ask AVL to automatically deflect the elevator in order to trim the airplane.
- The following commands will trim the airplane for 5 degrees of angle of attack:
 - load test.avl
 - oper
 - a a 5  This sets the angle of attack
 - d3 pm 0  This adjusts the elevator (d3) to set pitch moment to zero
 - x
- Now you can read the values CLtot and CDind on the screen. You can also see the elevator deflection.

Trimmed polar

- Run these commands for at least three other angles of attack and annotate the values of C_{Ltot} and C_{Dind} and elevator deflection. One of these angles should be at C_{Lmax} .
- Now you can fit a parabola to compute induced drag for any lift coefficient:

C_L	C_{Dind}
----	----
----	----
C_{Lmax}	----



$$C_{Dind} = a + bC_L + cC_L^2$$

Things you can do with AVL

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Control surface deflections

- As AVL does not detect separation, you may see high deflections on the control surfaces.
- The control surface usually stalls and loses effectiveness for deflections around 20 degrees. Therefore, check if the deflections of your airplane are less than this threshold.
- You already got the elevator deflections when doing the trimmed drag polar. The most critical case (highest deflection) occurs at C_{Lmax} .

Control surface deflections

- You can also get the elevator deflection at the maximum C_L condition with the following commands (assume $C_{L_{\max}}=1.8$):

- load test.avl

- oper

- a c 1.8

- d3 pm 0

- x

- (now you can read the elevator deflection on the screen).

This sets the desired lift coefficient

This adjusts the elevator (d3) to set pitch moment to zero

Control surface deflections

- It is also important to check if the rudder can generate enough moment to balance the airplane at one-engine out condition. Assume that you need a yawing moment of $C_n=0.014$:

➤ load test.avl

➤ oper

➤ a a 0

➤ d4 ym 0.014

➤ x

➤ (now you can read the rudder deflection on the screen).

This sets the angle of attack

This adjusts the rudder (d4) to set the desired yawing moment

Things you can do with AVL

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Exporting lift distribution

- You can export the lift distribution to a text file with the following commands:

➤ load test.avl

➤ oper

➤ o —————> This opens the options menu

➤ p —————> This opens the printing options

➤ t t t f —————> Here we ask to export total forces, surface forces, and strip forces

➤ <enter>

➤ a a 5 —————> This sets an arbitrary angle of attack

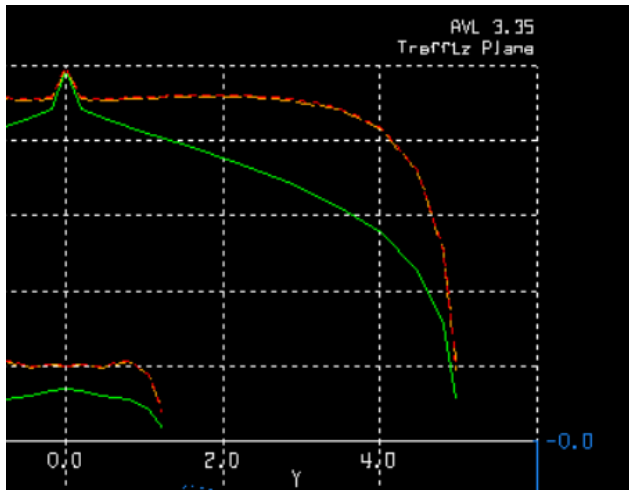
➤ x

➤ w

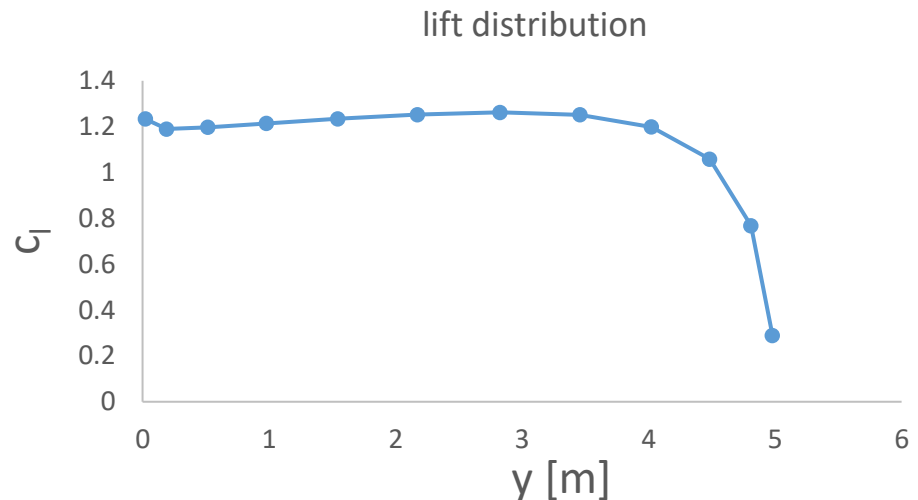
➤ results.txt —————> Name of the output file

Exporting lift distribution

- Then you can read the output file externally (Python, Matlab, or Excel) to generate plots.



VS



- You can also use this functionality to integrate AVL in your design framework. (But note that for efficient design frameworks, one should do this integration directly through memory rather than file I/O).

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Stability derivatives

- The stability derivatives can be obtained with:

➤ load test.avl

➤ oper

➤ a c 0.5

➤ st  Stability derivatives command

➤ st.txt  Name of the output file

		alpha		beta													
		-----		-----													
z' force CL	CLa =	5.529911	CLb =	0.000001													
y force CY	CYa =	-0.000005	CYb =	-0.340522													
x' mom. CL'	CLa =	-0.000003	CLb =	-0.291265													
y mom. Cm	Cma =	-3.813009	Cmb =	0.000000													
z' mom. Cn'	Cna =	0.000005	Cnb =	0.046403													
		roll rate p'		pitch rate q'													
		-----		-----													
z' force CL	CLp =	0.000002	CLq =	12.194499	CLr =	-0.000001											
y force CY	CYp =	-0.322705	CYq =	-0.000006	CYr =	0.260243											
x' mom. CL'	CLp =	-0.634842	CLq =	-0.000003	CLr =	0.214030											
y mom. Cm	Cmp =	0.000002	Cmq =	-23.596344	Cmr =	-0.000000											
z' mom. Cn'	Cnp =	-0.033034	Cnq =	0.000006	Cnr =	-0.058045											
		flap		d1		aileron		d2		elevator		d3		rudder		d4	
		-----		-----		-----		-----		-----		-----		-----		-----	
z' force CL	CLd1 =	0.047243	CLd2 =	-0.000000	CLd3 =	0.007839	CLd4 =	0.000000									
y force CY	CYd1 =	-0.000000	CYd2 =	0.007494	CYd3 =	-0.000000	CYd4 =	-0.002374									
x' mom. CL'	CLd1 =	-0.000000	CLd2 =	0.009876	CLd3 =	-0.000000	CLd4 =	-0.000087									
y mom. Cm	Cmd1 =	-0.019128	Cmd2 =	-0.000000	Cmd3 =	-0.028604	Cmd4 =	0.000000									
z' mom. Cn'	Cnd1 =	0.000000	Cnd2 =	-0.000967	Cnd3 =	0.000000	Cnd4 =	0.000906									
Trefftz drag	CDffd1 =	0.001356	CDffd2 =	-0.000000	CDffd3 =	-0.000075	CDffd4 =	-0.000000									
span eff.	ed1 =	0.014491	ed2 =	0.000000	ed3 =	0.032556	ed4 =	0.000000									

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